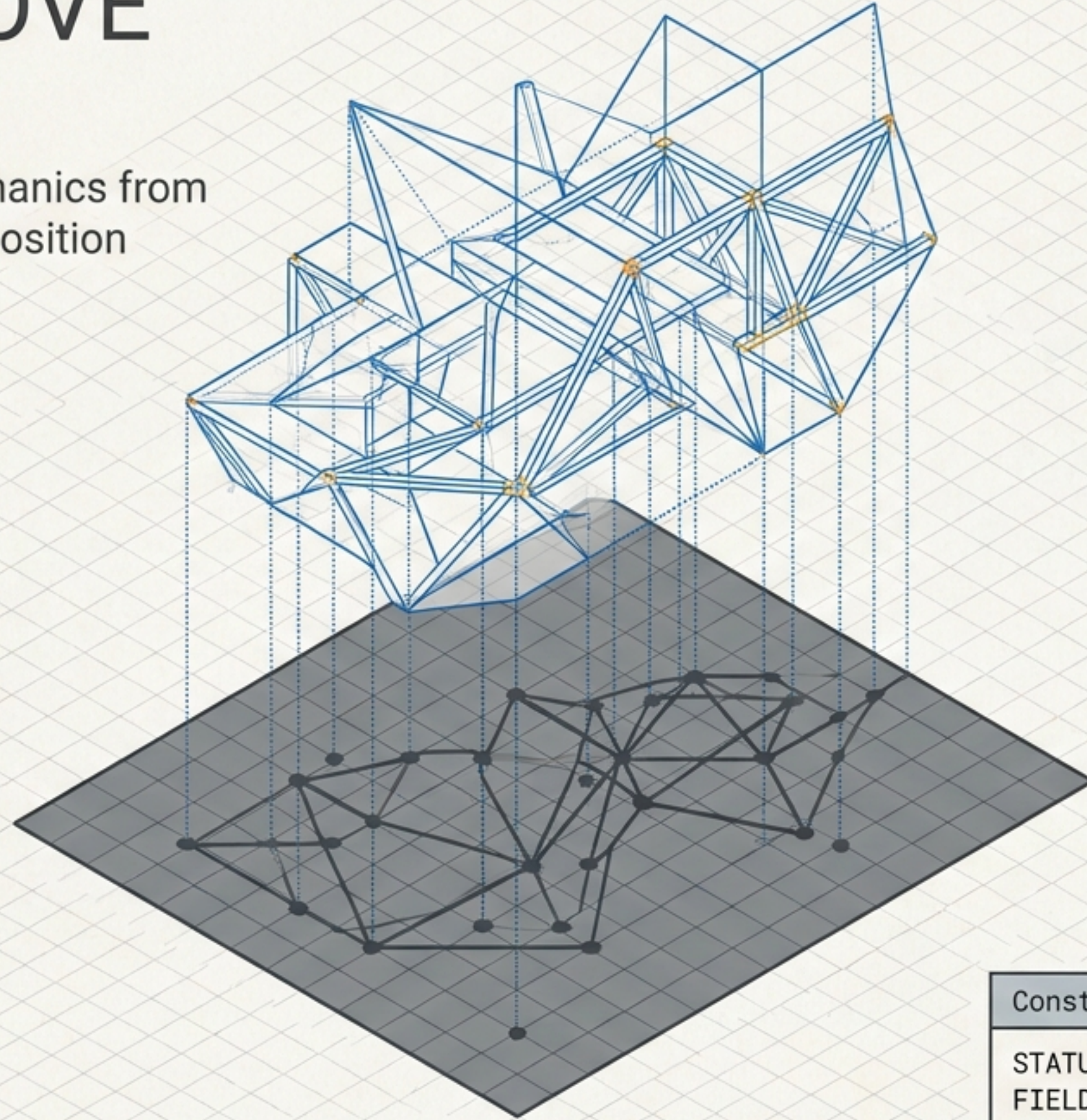


THE LIFT ABOVE THE RECORD

Reconstructing Quantum Mechanics from
Admissible Probabilistic Composition

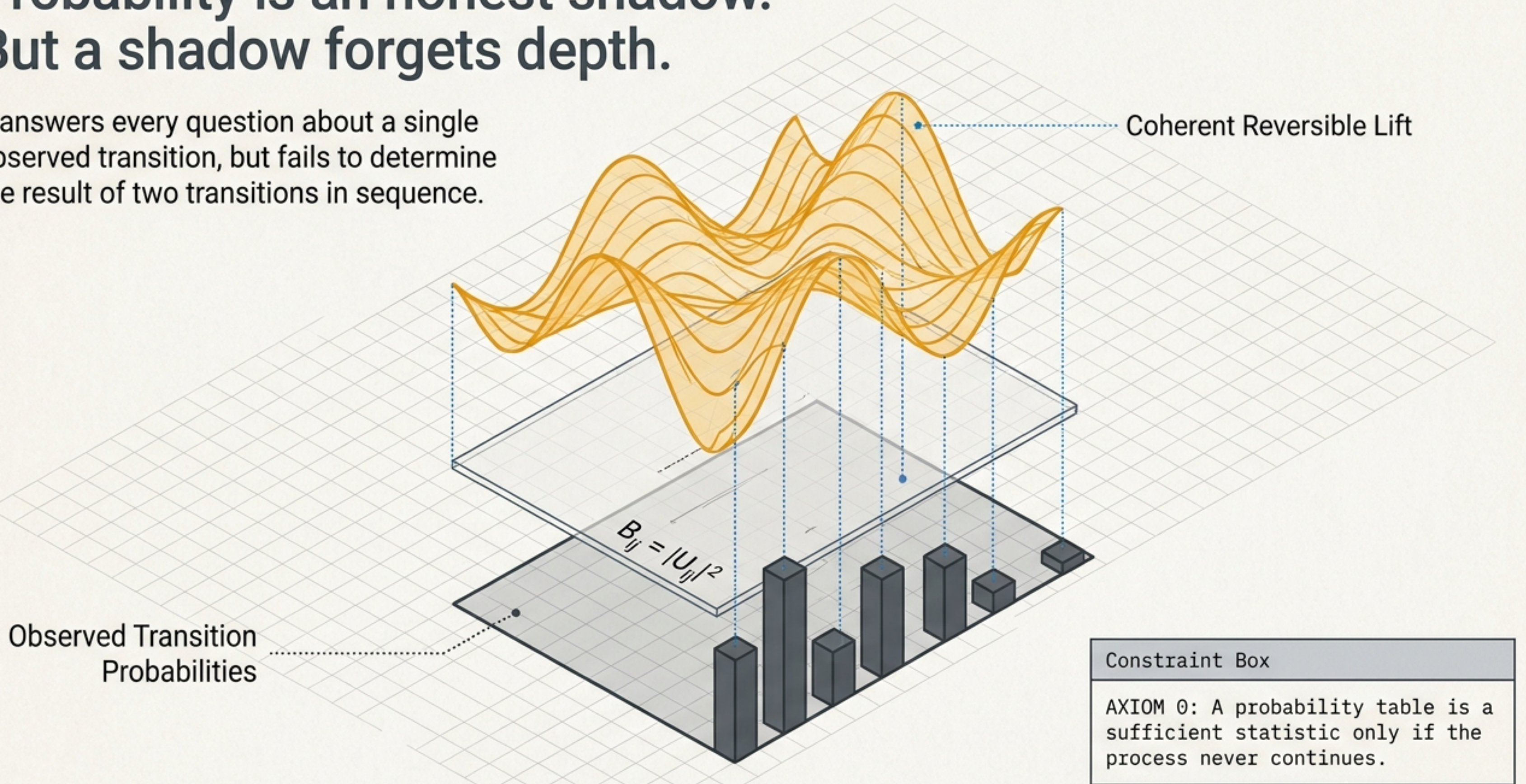


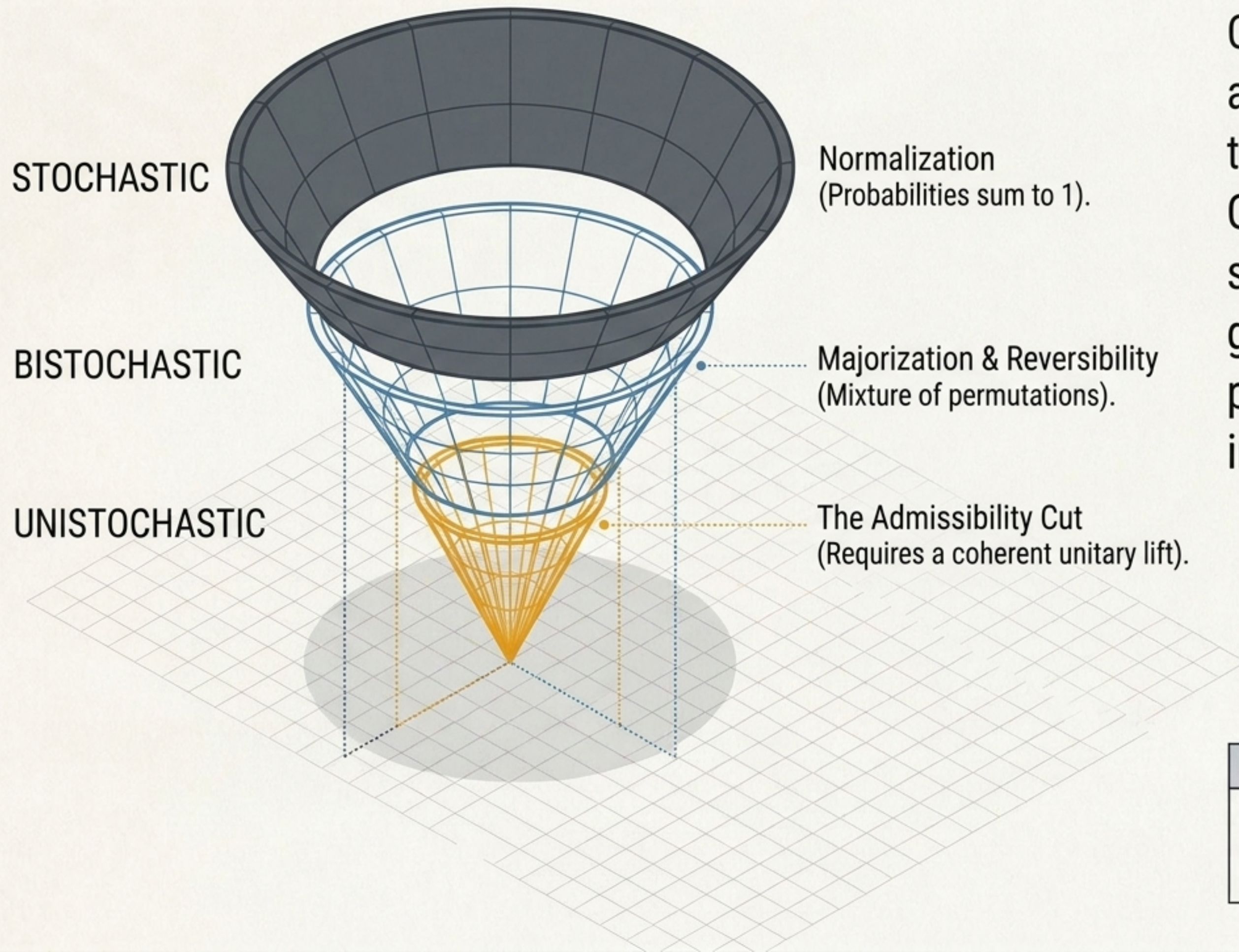
Constraint Box

STATUS: System Audit Initiated //
FIELD: Foundations of Quantum Theory

Probability is an honest shadow. But a shadow forgets depth.

It answers every question about a single observed transition, but fails to determine the result of two transitions in sequence.





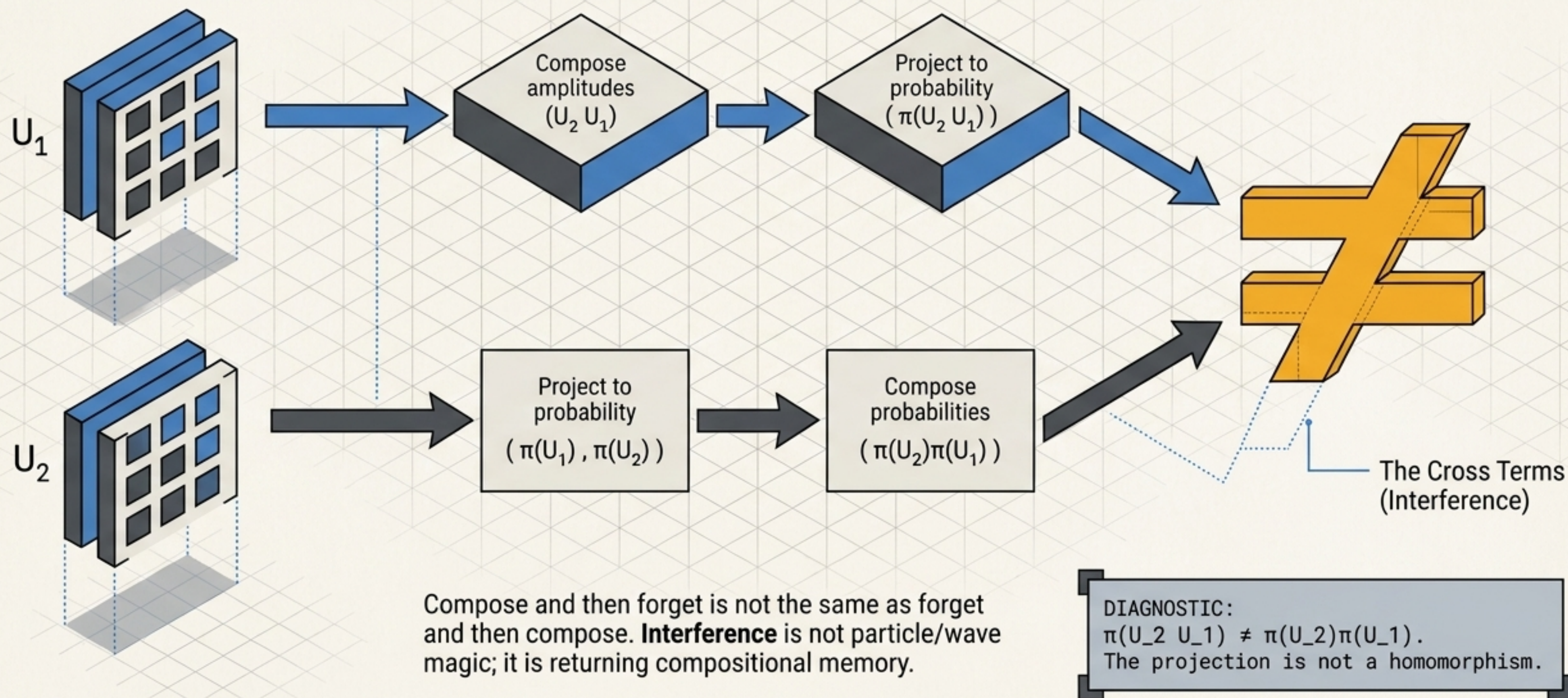
Classical probability studies all legal movements through the simplex. Quantum mechanics studies only the restricted geometry of transitions that preserve relational information.

Constraint Box

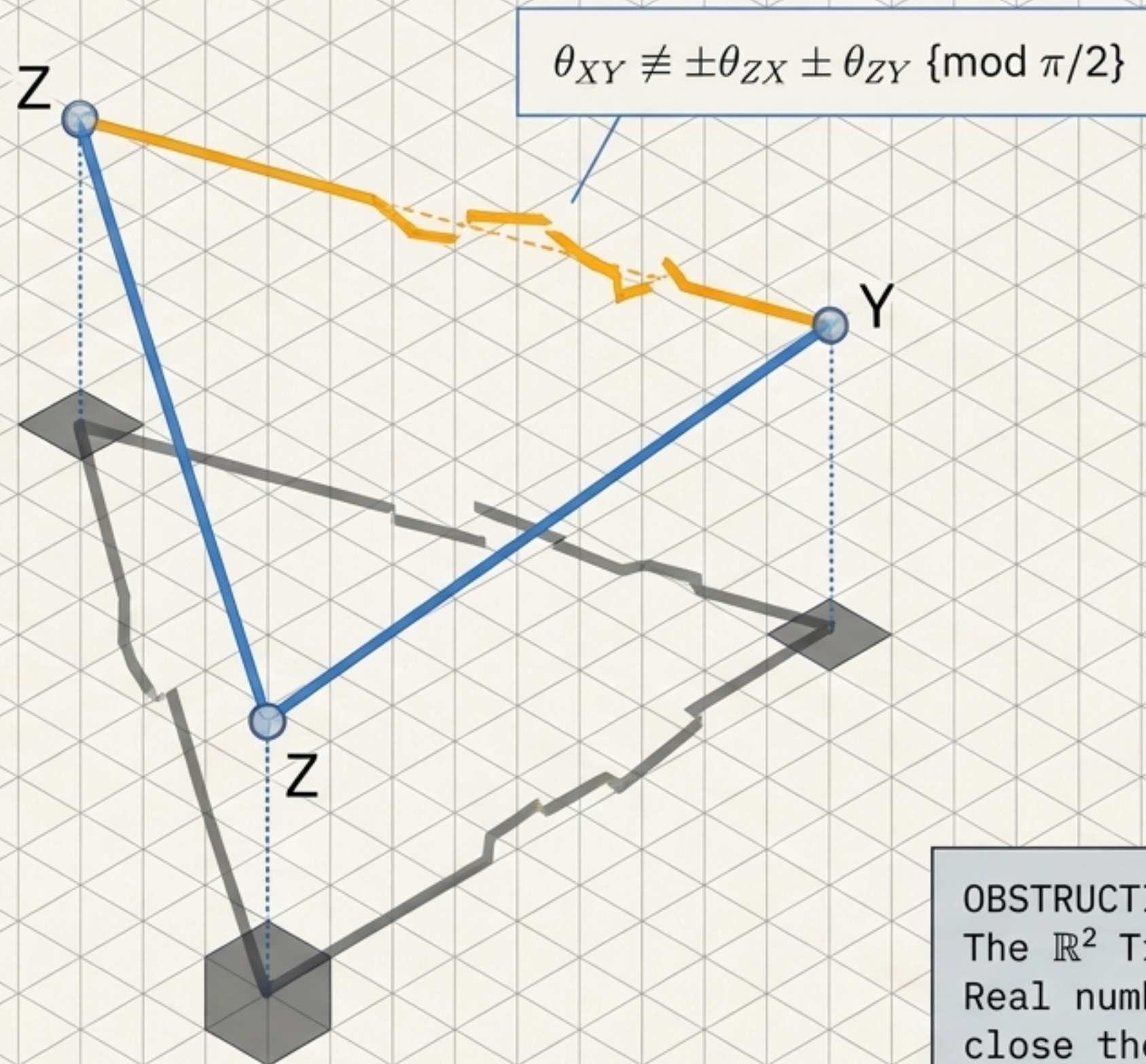
CONSTRAINT 1: Probability is admitted only when a coherent reversible lift exists above it.

Interference is Memory: The Non-Commutation of Composition and Forgetfulness

Synthesis: Compose and then forget is not the same as forget and then compose. Interference is not particle/wave magic; it is returning compositional memory.

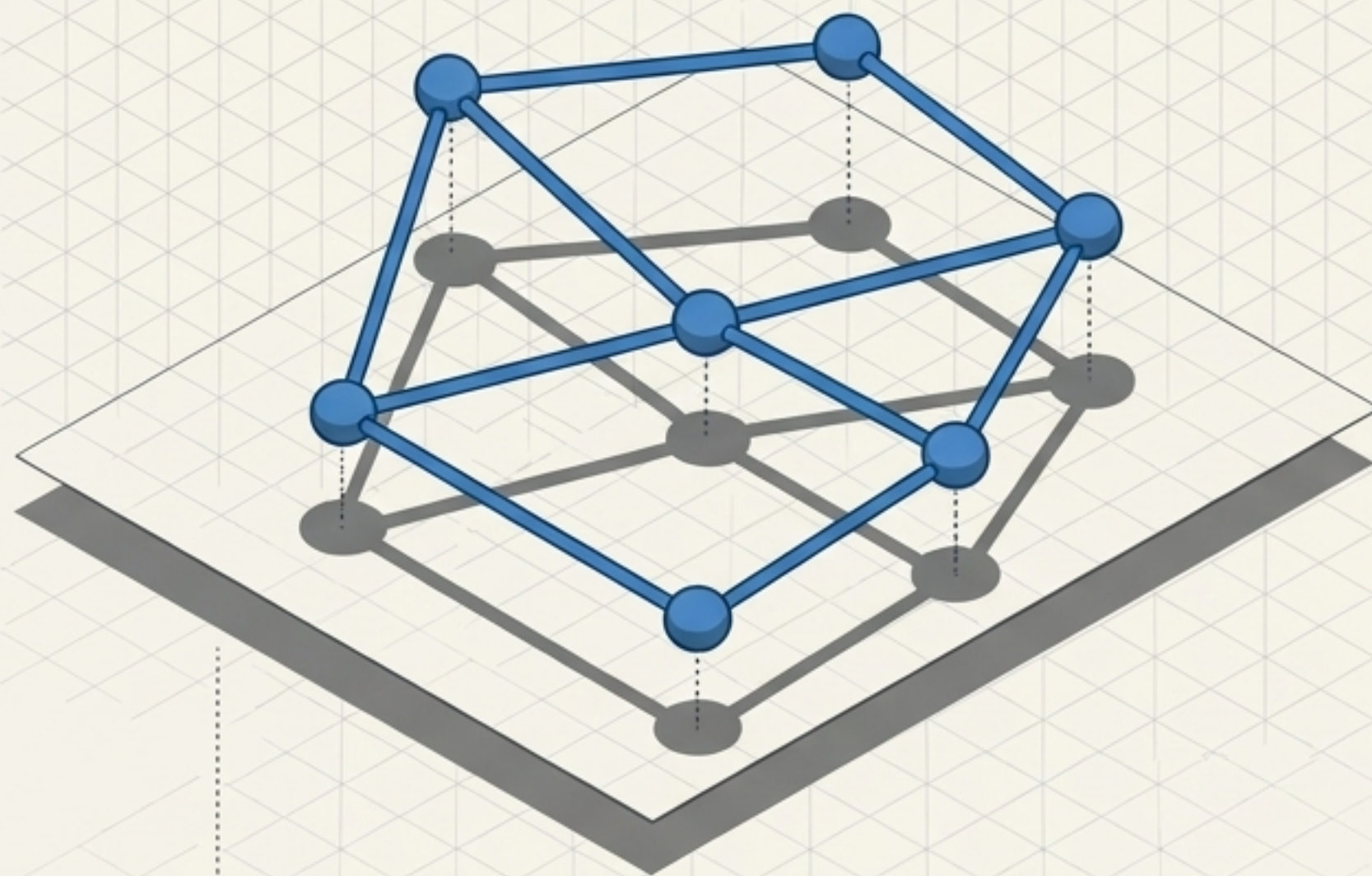


Three mutually unbiased bases (MUBs) can exist pairwise in $\mathbb{R}^2$, but all three cannot be globally glued. Local edgewise admissibility does not inherit global consistency.

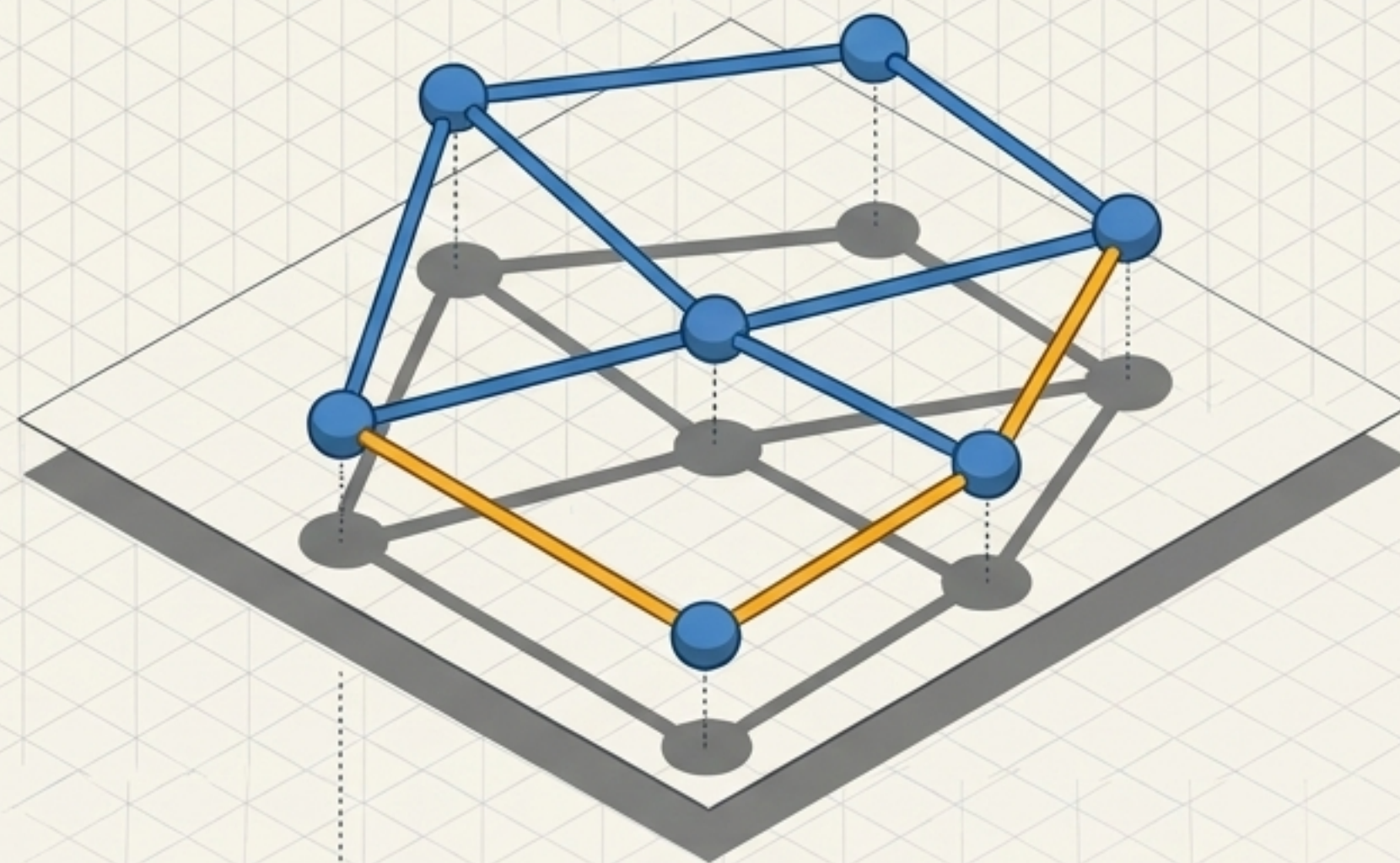


OBSTRUCTION:
The $\mathbb{R}^2$ Triangle.
Real numbers fail to
close the minimal cycle.

A cycle demands a second account of something already assigned. The required coordination is a codimension-one event. Under continuous measure, the probability of real realizability drops to zero the moment $\beta_1 \geq 1$.



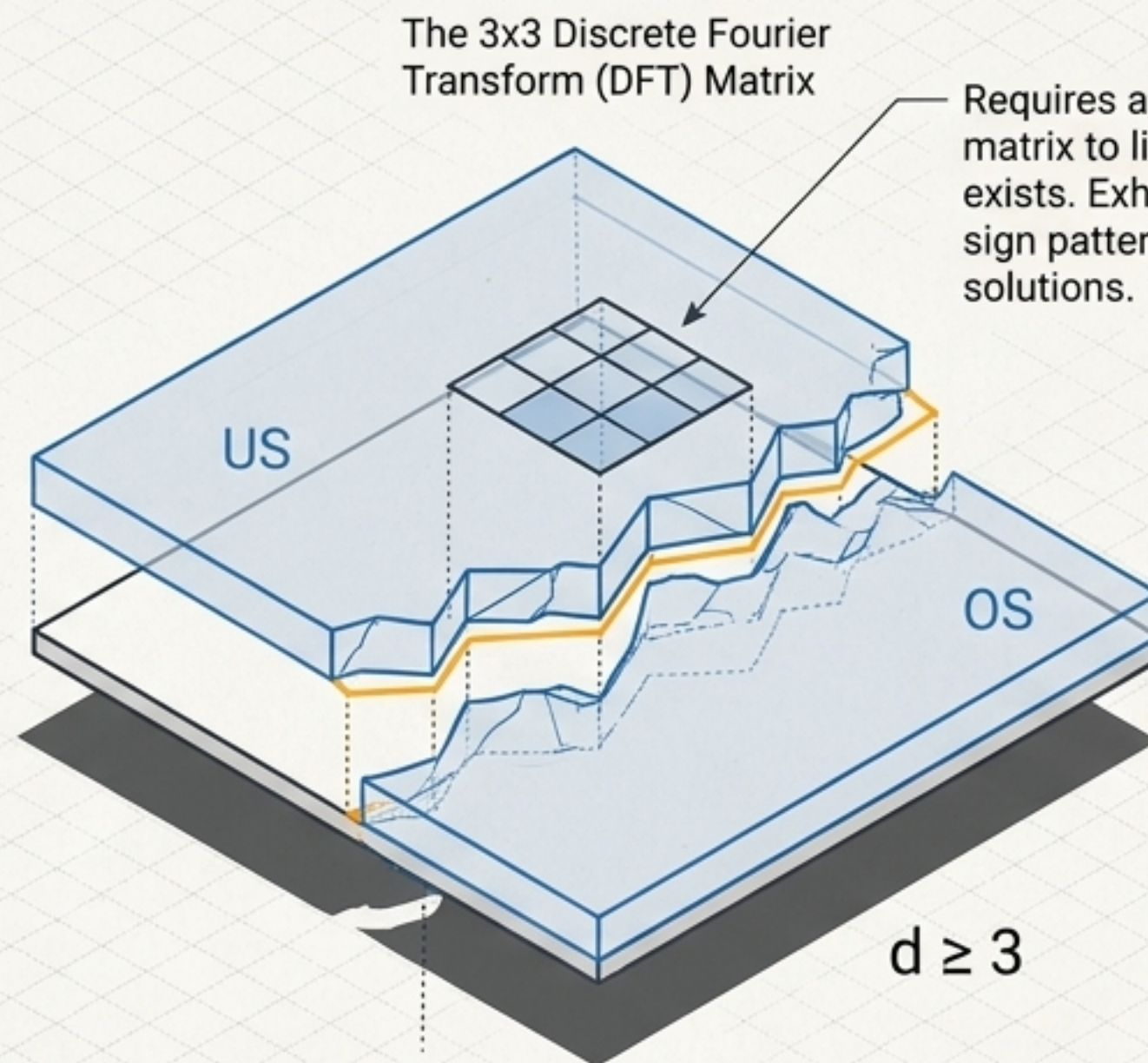
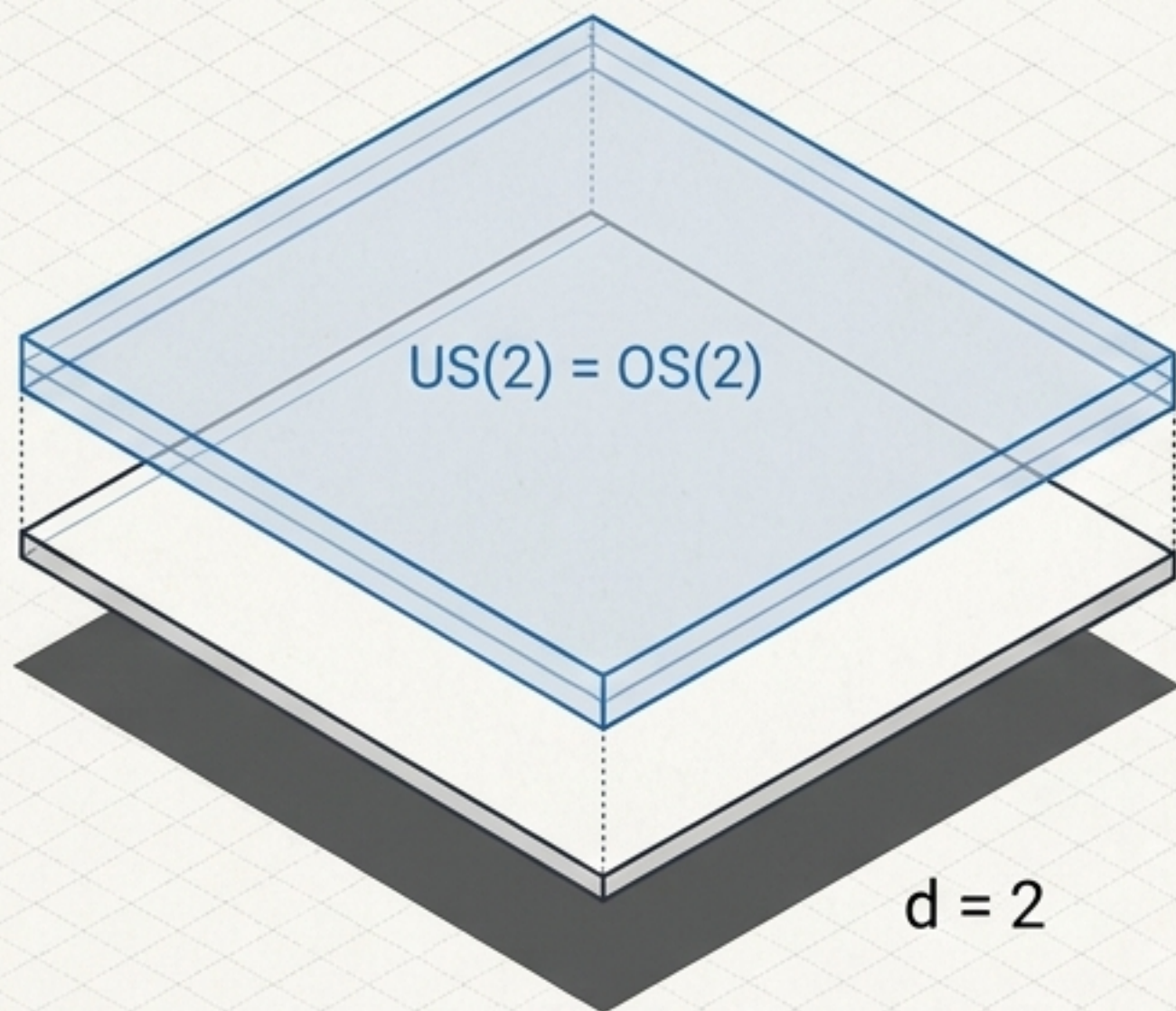
The Spanning Tree:
Paths are unique. Any local lift propagates globally without contradiction.



The Closed Cycle:
The Cycle Rank
 $\beta_1(G) = |E| - |V| + 1$.

THEOREM:
Global $\mathbb{R}$ -realizability
holds if and only if the
context graph is a forest.

At $d=2$, cycles are needed to break real numbers. From $d=3$ onward, a single edge is sufficient to separate $\mathbb{R}$ from $\mathbb{C}$.



The 3x3 Discrete Fourier Transform (DFT) Matrix

Requires a 3x3 real Hadamard matrix to lift orthogonally. None exists. Exhaustive check of 512 sign patterns yields zero solutions.

Constraint Box

THEOREM ($n \geq 3$ Incompleteness):
 $OS(3) \subset US(3)$.
 Single-edge failure.

How must independent systems compose? If we assume a joint state should be exactly as informative as local tomography on its parts, the scalar field is uniquely constrained.



$\mathbb{R}$ (Real)

Right pan is heavier
(hidden joint
parameters).



$\mathbb{H}$ (Quaternionic)

Left pan is heavier
(naive product is
over-complete).



$\mathbb{C}$ (Complex)

Perfect balance.

Local Measurements A
+
Local Measurements B

The Joint State

Constraint Box

ASSUMPTION 1 (Tested, not derived):
Local Tomography
 $\beta(d_A d_B) = \beta(d_A) \beta(d_B).$

Synthesis: Complex Hermitian state spaces are the unique mathematical environment where the tensor product construction perfectly realizes local tomography **without missing** or overdetermined parameters.

The Scalar Field Diagnostic Table

	Real ($\mathbb{R}$)	Complex ($\mathbb{C}$)	Quaternionic ($\mathbb{H}$)
Dimension Counting $d_A=d_B=2$	10 vs 9 (Under-counts)	16 vs 16 (Exact match)	28 vs 36 (Over-counts)
Missing/Extra Data	Hides non-local correlational structure	Spanned entirely by local tensor products	Products satisfy nontrivial linear relations
MUB Upper Bound	$\leq d/2 + 1$	$\leq d + 1$	$\leq 2d + 1$

Constraint Box

DIAGNOSTIC: $\mathbb{C}$ is the unique fixed point for local tomography under the Kronecker product.

Locally inadmissible
(Antisymmetric real matrix)

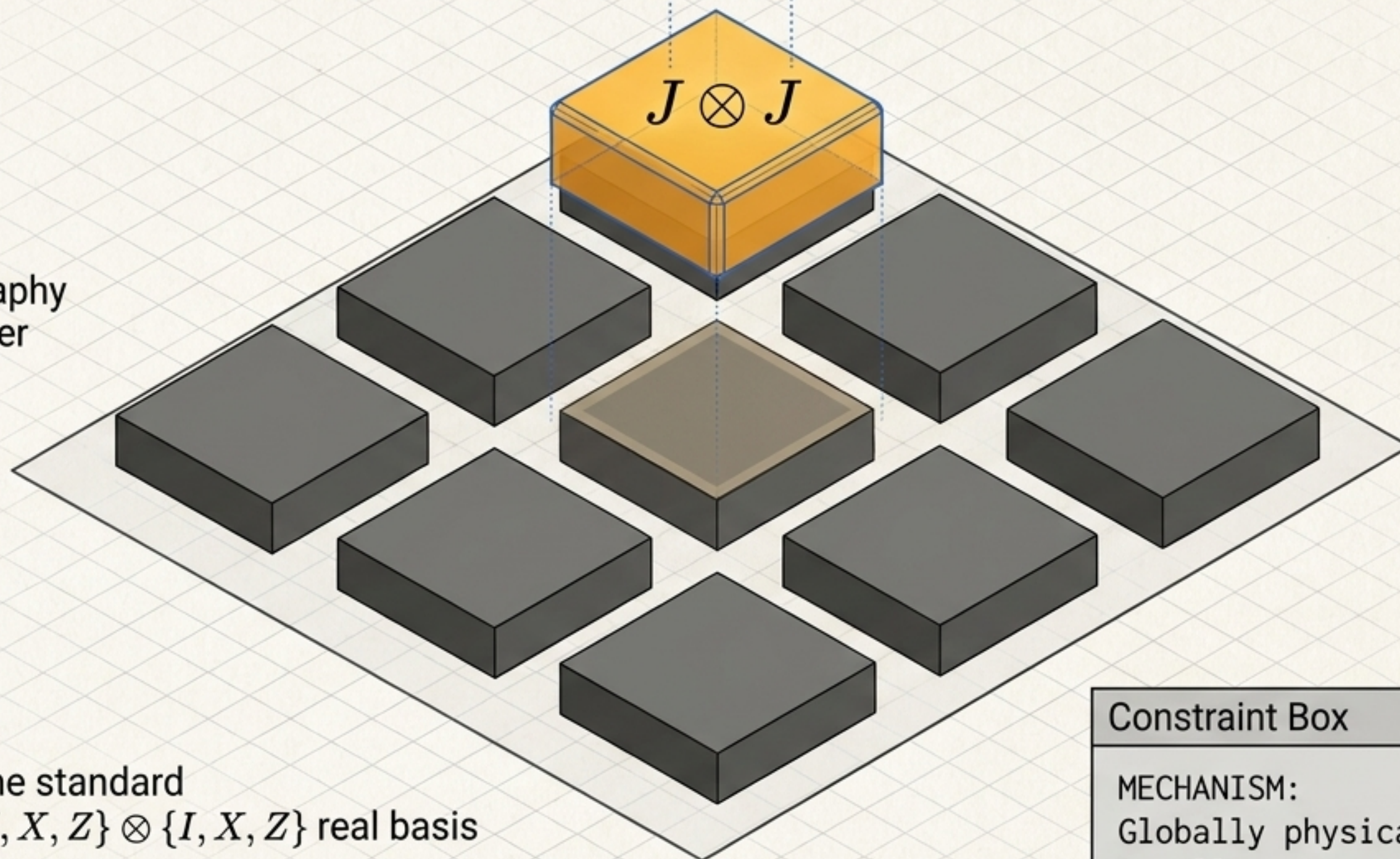
Globally admissible
(Symmetric real matrix)

Real composite systems carry
a genuinely nonlocal degree of
freedom.

The exact axis responsible for
the failure of real local tomography
is a bounded physical parameter
($|c| \leq 1$) that vanishes entirely
under partial trace.

It is the real entanglement
witness.

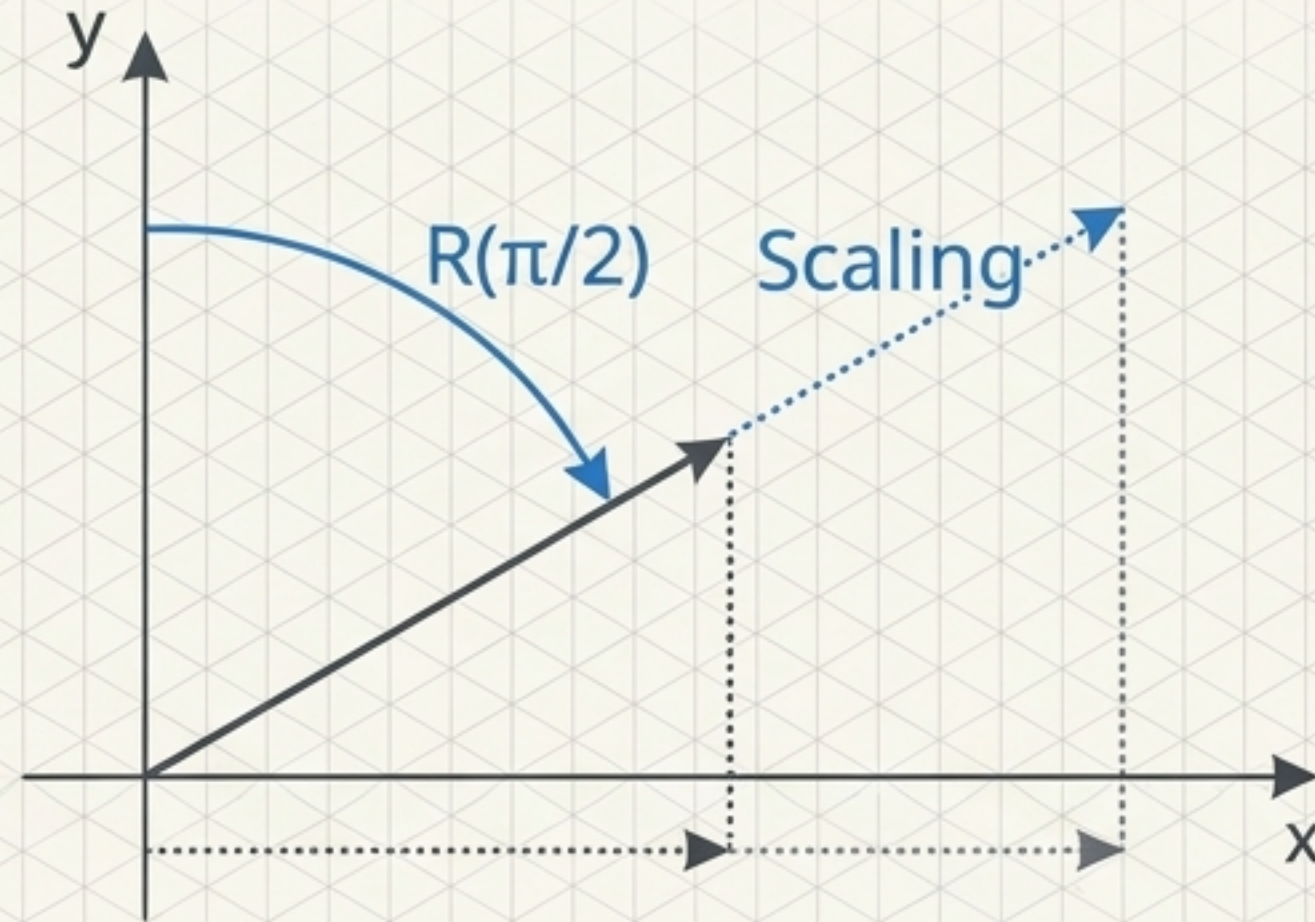
The standard
 $\{I, X, Z\} \otimes \{I, X, Z\}$ real basis



Constraint Box

MECHANISM:
Globally physical $\neq$
Locally recoverable.
The invisible dimension.

$$\hat{J} = R(\pi/2) \equiv \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \equiv i$$

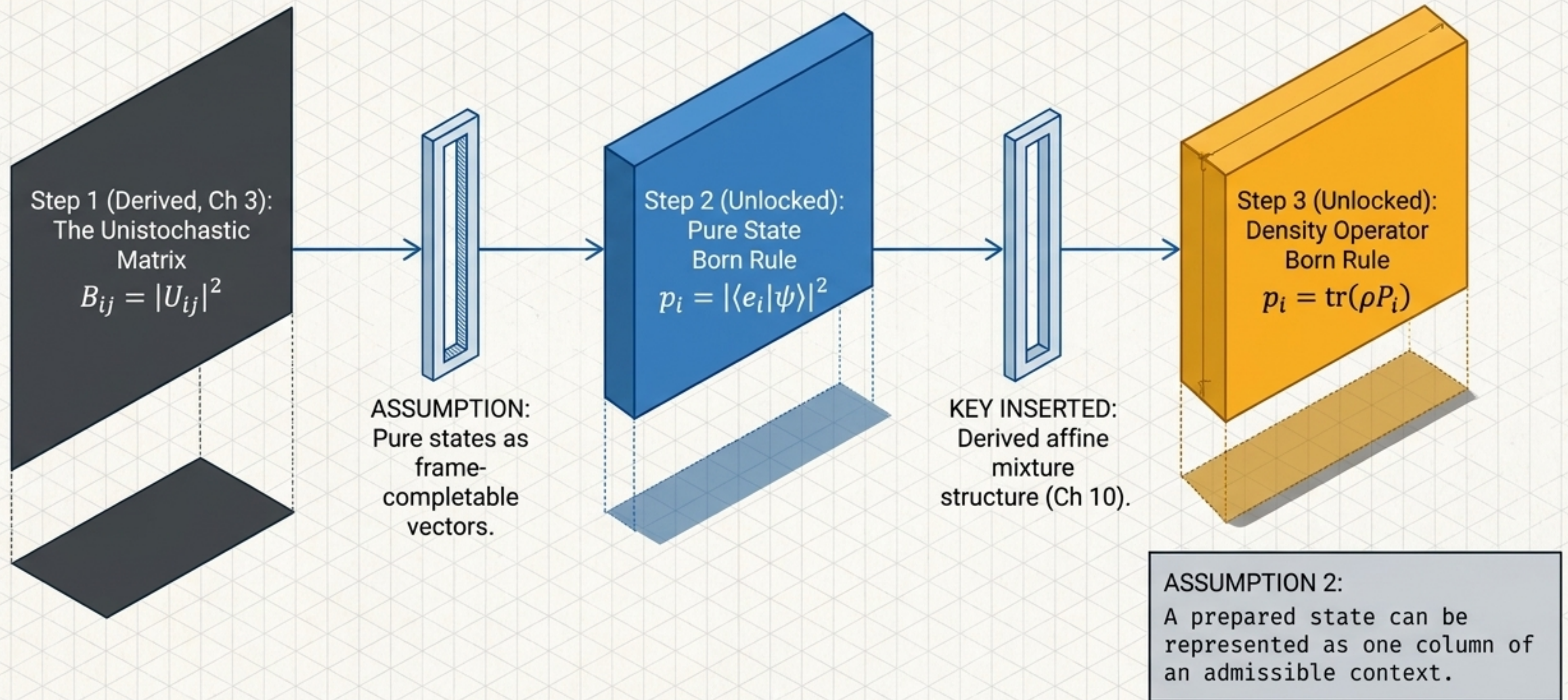


The Amplitwist. Complex numbers were never a spooky physical postulate. They are simply the geometry of simultaneous scaling and quarter-turn rotation ($\hat{J}$). This exact geometry is required to prevent the phase-redistribution paradox in composite systems.

Constraint Box

SYNTHESIS: Complex multiplication is exactly amplitwist composition.
 $\hat{J}^2 = -I$.

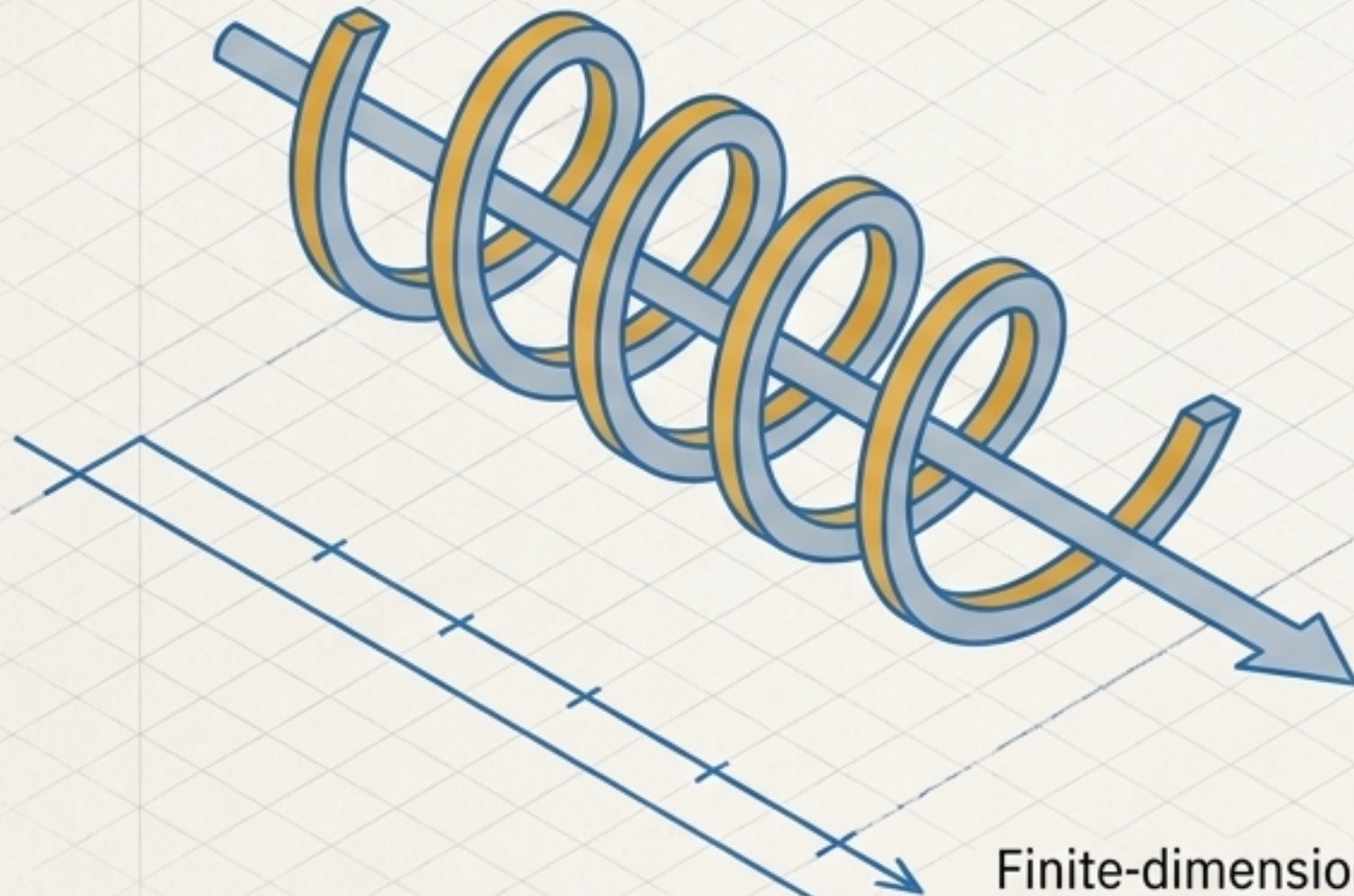
The **Born rule** is not an independent postulate. It is literally one column of the the already-established context-transition identity, unlocked by a single operational assumption about state preparation.



Schrödinger's equation is merely the infinitesimal form of continuous reversible composition.
POVMs are not new physics; they are sharp projective measurements viewed from a subsystem.

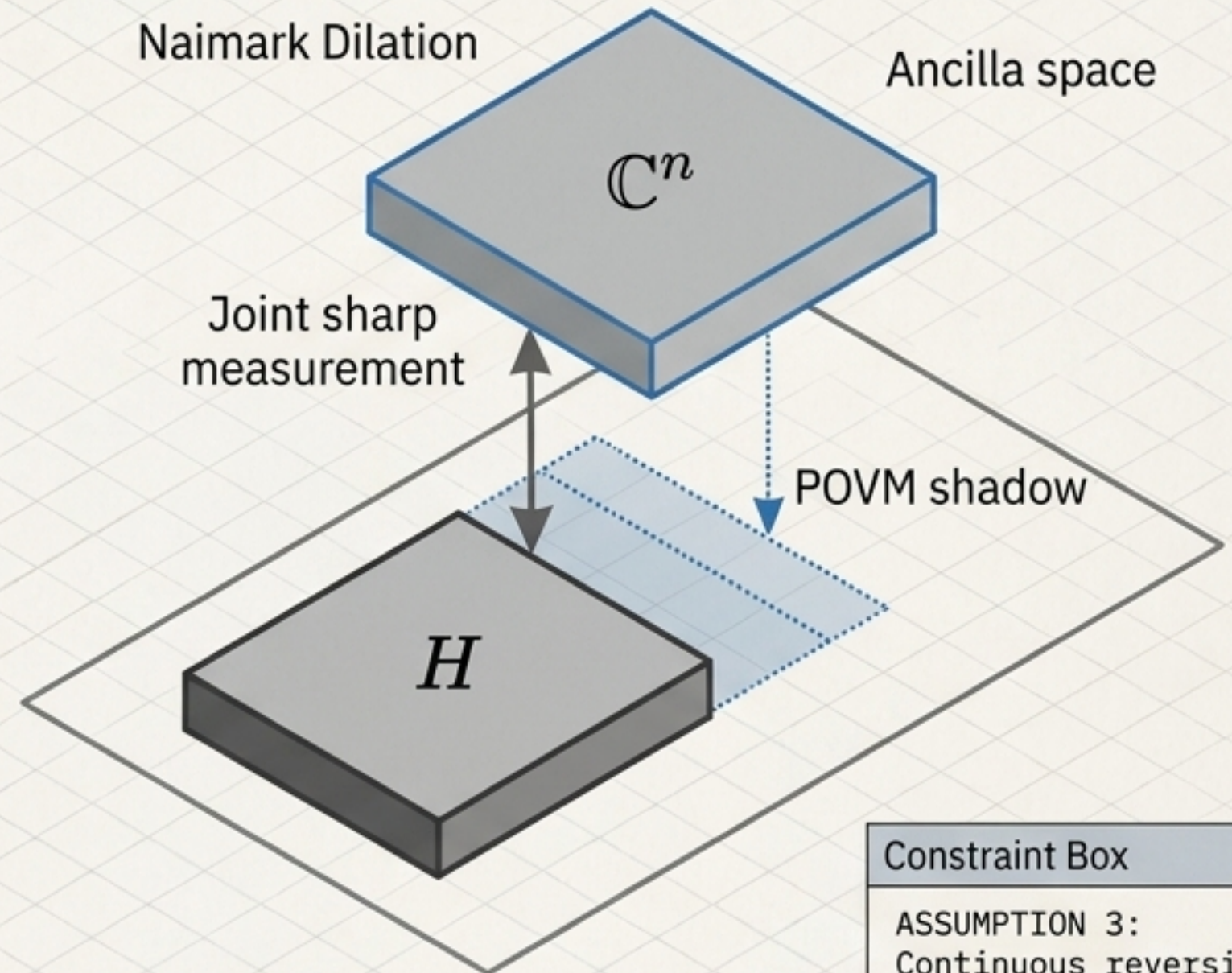
Dynamics

$$U(t) = e^{-itH/\hbar} \rightarrow i\hbar\partial_t|\psi\rangle = H|\psi\rangle$$



Finite-dimensional
generator theorem.

Measurement



Constraint Box

ASSUMPTION 3:
Continuous reversible
time evolution.
(Measurement outcomes
as $\mathbb{R}$ labels).

Quantum mechanics is not an arbitrary collection of paradoxes. It is a rigid architectural response to exactly four operational assumptions applied to admissible probability.

DERIVED

Probability simplex
Entropy non-decrease
 $n \geq 3$ incompleteness
Tensor product spanning
Positivity
Partial trace
Real entanglement witness
Naimark dilation

REFORMULATED

Minimal graph witnesses
Amplitwist equivalence

ASSUMED (The 4 Final Costs)

1. Local Tomography.
2. States as frame-completable vectors.
3. Real-valued outcome labels.
4. Continuous unitary time evolution.

UNRESOLVED

Density thresholds
 $d > 2$ composite entanglement
physical identification of H as energy

Constraint Box

AUDIT COMPLETE:
4 Assumptions.
No hidden postulates.
The architecture holds.